Poisson counts vs exponential waiting times

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This note clarifies a point I mis-stated during Monday's lecture. Read S. Coles [1] §4.3.3 for more details and derivations.

Event arrival rates

On Monday we wrote that extremes (where $x > u$) arrive at rate λ per year and assumed these events to be independent and stationary. The GPD gives the probability distribution for the extremes, *conditional on them being extreme*. To say anything about return periods, we *also* need a model for their arrival.

There are two questions, which are (as we will see) closely related.

How long until the next exceedance?

The waiting time W between consecutive cluster maxima is frequently modeled **exponential** (not Poisson as I mistakenly said on Monday).

$$P(W > w) = e^{-\lambda w}$$

where w is measured in years (the same units as λ). The mean waiting time is $1/\lambda$ years (so λ has units of year^{-1}.) The exponential distribution is continuous: it models durations.

> **i** Note
>
> Assuming exponential waiting times is a convenient and practical assumption widely used in the literature. However, it is a modeling choice / assumption rather than a theoretical result. As we noted, weather and climate extremes rarely meet this assumption (the expected waiting time between hurricanes is quite different between December vs June, or an El Niño vs La Niña year).

How many exceedances in a year?

If the waiting times are exponential, the number of exceedances in a fixed interval follows a Poisson distribution. The count N in one year is Poisson with rate λ:

$$P(N = k) = \frac{\lambda^k}{k!} e^{-\lambda}$$

> **Note**
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> Once again, the Poisson distribution for the number of counts in a season is common but not required. Other distributions, like the Negative Binomial, could also be used, although this would somewhat complicate the return period calculations. It is also fairly common to see applications that use the Poisson model for counts per year but not the exponential waiting period model, for example to model tropical cyclone arrivals in a given location which occur only during the summer months.

The return level formula

The compound-Poisson model combines the Poisson rate λ with the GPD to give the T-year return level:

$$x_T = u + \frac{\sigma}{\xi}\left[(T\lambda)^{\xi} - 1\right]$$

for $\xi \neq 0$. When $\xi = 0$, the formula becomes $x_T = u + \sigma \ln(T\lambda)$.

Bibliography

[1] S. Coles, *An Introduction to Statistical Modeling of Extreme Values*. in Springer Series in Statistics. London: Springer, 2001.